26-BS-BI201 — Biology for Engineers

Course Code	26-BS-BI201	**Category**	Basic Science Course
Course Title	Biology for Engineers	**L – T – P**	2 – 0 – 0
Credits	2	**Contact Hours**	26
Semester	Second	**Assessment**	Theory: CA 30 + ESE 70

Prerequisites

A two-credit course scoped at about twenty-six hours (13 h per credit). It is self-contained and assumes only the general science of Class X; a student without Higher Secondary Biology is fully provided for. Assumed:

- The general biology and chemistry of Class X (WBBSE / CBSE / ICSE Science).
- No Higher Secondary Biology is assumed; Module I supplies what such a student requires.
- Elementary chemistry to the level of the concurrent or preceding Chemistry course.

Course Objectives

- To present biology to the engineer as a source of design principles rather than as descriptive natural history.
- To establish the molecule, the cell and the gene as the units of living systems.
- To introduce the bio-engineering interfaces — biomechanics, biomaterials, sensors and biotechnology.
- To connect biological principle to engineering application across the branches.

Course Outcomes

On completion of the course the student will be able to:

CO	Statement	Bloom
CO1	Explain the molecular and cellular basis of living systems and the flow of genetic information.	L2 – Understand
CO2	Relate the structure and properties of biomolecules and biomaterials to their engineering use.	L3 – Apply
CO3	Explain the bio-engineering interfaces — biomechanics, biosensors and biotechnology — and their applications.	L2 – Understand
CO4	Draw an engineering analogy or design principle from a biological system.	L4 – Analyse

CO–PO Articulation Matrix

Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	1	1	1	–	–	–	1
CO2	3	2	2	1	1	1	2	–	–	–	1
CO3	2	2	2	1	2	2	2	–	–	–	2
CO4	2	3	3	2	2	2	2	–	–	–	2

Course Plan — Module-wise Syllabus

Module	Title	Contents	Hours
I	The Cell and the Molecules of Life	Life as a chemical system; the cell as the structural and functional unit; prokaryotic and eukaryotic cells; the biomolecules — carbohydrates, lipids, proteins and nucleic acids — and their structure and function; enzymes as catalysts; bioenergetics in outline.	7
II	Genetics and the Flow of Information	The gene and the chromosome; the structure of DNA and RNA; replication, transcription and translation in outline; the genetic code; the elements of inheritance; an introduction to genetic engineering and its applications.	6
III	Biomaterials, Biomechanics and Bio-inspired Design	Natural materials — bone, wood, silk, shell — and their remarkable properties; biomaterials and biocompatibility in outline; the mechanics of the musculoskeletal system; bio-inspired and biomimetic design with engineering examples.	7
IV	Biotechnology, Biosensors and Applications	The elements of biotechnology and its industrial, agricultural and medical applications; the biosensor — principle and examples; bioinformatics in brief; the interface of biology with the engineering disciplines, including biomedical instrumentation and environmental biotechnology.	6

Cross-cutting notes. The course is deliberately application-facing: each module ends with an engineering analogy or design principle, so that the biology serves the engineer. It connects to the environmental and effluent content of the later years and, for the relevant branches, to biomedical and bioprocess electives.

Text and Reference Books

- Annamma Odaneth, Biology for Engineers, Khanna Publishing House (AICTE Prescribed Textbook).
- Nelson and Cox, Lehninger Principles of Biochemistry (selected chapters), Macmillan.
- Uma Devi Koduru, General Biology, Khanna Publishing House.
- A. K. De, Biology for Engineers, university-level text.
- S. B. Primrose, Principles of Gene Manipulation (selected chapters), Blackwell.
- T. A. Brown, Biotechnology (introductory chapters).

Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Biology for Engineers / Introductory Biology (IITs), 8 / 12 weeks.
- NPTEL — Bioengineering: An Interface with Biology and Medicine, 12 weeks.
- SWAYAM — Biology for Engineers, as offered.

Question-paper note. The paper is to test understanding and application rather than recall of biological detail. At least one Group C question is to ask the student to draw an engineering design principle from a biological system.